

Theorem 4. The synchronous breathing of the ring

The mean-transport identity and the exact closure of the rosettes $T_c = 2\pi / \nu$

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The cycloring research program · v1.0.0 · Theorem Monograph Edition

Study	CYCLORING
Study	CYCLORING — the Gamma-periods, the defect chain, and the synchronous breathing of a vortex ring
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Abstract

The monograph proves the central dynamical statement of the program. The radial breathing program $r(t) = R_0(1 + \epsilon \cos \nu t)$ with the rigid $2\pi/N$ choreography is not an orbit of the Kirchhoff flow — the polygon has no radial velocity of its own (the pair cancellations); the radial drive is supplied by a pump. The angular transport of the ring, however, is not prescribed: it follows the frozen law $\Lambda_0/r(t)^2$ with the adiabatic invariant $\Lambda_0 = \Gamma(N-1)/(4\pi)$.

The core of the theorem is the integral mean-transport identity: the cycle mean of $(1 + \epsilon \cos u)^{-2}$ equals $(1 - \epsilon^2)^{-3/2}$. The synchronous choice of the frequency $\nu = \omega_L(R_0)(1 - \epsilon^2)^{-3/2}$ makes the phase accumulated over a cycle exactly 2π : every rosette $z_k(t) = r(t)e^{i(\nu t + 2\pi k/N)}$ is closed, and the configuration returns at $T_c = 2\pi/\nu$. Both statements are exact, not approximations.

Certificates: W5 — the mpmath residual of the identity $1.0e-30$ over the grid ϵ in $[0.05, 0.6]$, the 2π phase advance with the error $3.6e-13$; W6 — the closure of the level rosettes $N = 7$ ($\epsilon = \epsilon_7$) with the residual $1.5e-15$ and the shape rigidity $1e-16$.

At a glance

Breathing program	$r(t) = R_0(1 + \epsilon \cos \nu t)$, $\theta_k = \nu t + 2\pi k/N$
Adiabatic invariant	$\Lambda_0 = \omega_L R^2 = \Gamma(N-1)/(4\pi)$ — radius-free
Transport identity	$M[(1 + \epsilon \cos u)^{-2}] = (1 - \epsilon^2)^{-3/2}$
Synchronous frequency	$\nu = \omega_L(R_0)(1 - \epsilon^2)^{-3/2}$
Phase advance per cycle	exactly 2π
Closure	$T_c = 2\pi/\nu$; $z_k(T_c) = z_k(0)$

1. The polygon invariance and the role of the pump

The pair cancellations of the Kirchhoff field on the regular polygon give: the total velocity of every vortex is tangential, $\dot{z}_k = i\omega_L(r) z_k$, and the radial component vanishes identically. Hence the family of frozen polygons is invariant in the flow — the free ring started from the modulated initial condition rotates rigidly at the radius $R_0(1 + \epsilon)$ and does not breathe. The breathing is an external radial program; its kinematic honesty is that the angular transport is not assigned but computed: $\dot{\phi}(t) = \Lambda_0/r(t)^2$.

$$\Lambda_0 = \omega_L R^2 = \frac{\Gamma(N-1)}{4\pi}, \quad \dot{\phi}(t) = \frac{\Lambda_0}{r(t)^2}$$

2. The mean-transport identity

The integral $\int_0^{2\pi} (du)/((1+\varepsilon \cos u)^2)$ is computed by $t = \tan(u/2)$: the base integral $\int_0^{2\pi} (du)/(a + b \cos u) = (2\pi)/(\sqrt{a^2-b^2})$ for $a > |b|$, and the differentiation in a gives $\int_0^{2\pi} (du)/((a+b \cos u)^2) = (2\pi a)/((a^2-b^2)^{3/2})$. At $a = 1$, $b = \varepsilon$ we obtain the identity.

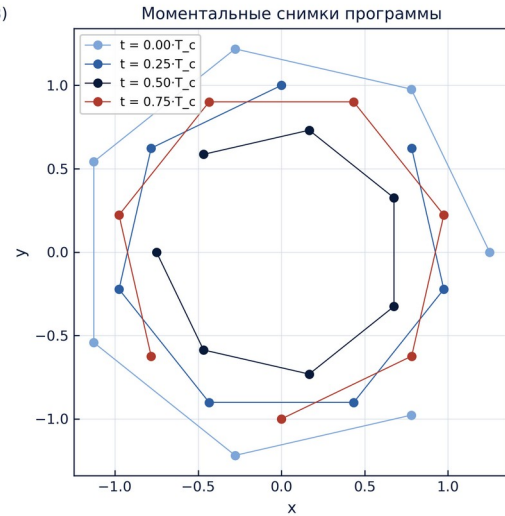
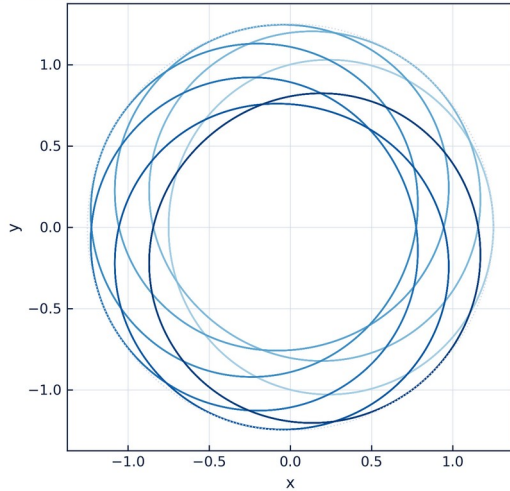
$$\frac{1}{2\pi} \int_0^{2\pi} \frac{du}{(1 + \varepsilon \cos u)^2} = \frac{1}{(1 - \varepsilon^2)^{3/2}}$$

3. The synchronous frequency and the closure

The phase over a cycle: $\int_0^{T_c} (\Lambda_0 dt)/(r(t)^2) = (\Lambda_0)/(\nu R_0^2) \cdot (2\pi)/((1-\varepsilon^2)^{3/2})$. The synchronous choice $\nu = (\Lambda_0/R_0^2)(1-\varepsilon^2)^{-3/2} = \omega_L(R_0)(1-\varepsilon^2)^{-3/2}$ cancels the fraction: the accumulated phase equals 2π exactly. Then $z_k(T_c) = z_k(0)$ for all k : the rosettes are closed, the configuration returns, and on the way every vortex rides the limason $r(\theta) = R_0(1+\varepsilon \cos(\theta - 2\pi k/N))$. The closure is a consequence of periodicity, not of the smallness of ε .

$$\int_0^{T_c} \frac{\Lambda_0 dt}{r(t)^2} = \frac{\Lambda_0}{\nu R_0^2} \cdot \frac{2\pi}{(1 - \varepsilon^2)^{3/2}} = 2\pi \quad (\nu = \omega_L(R_0)(1 - \varepsilon^2)^{-3/2})$$

Розетки синхронного дыхания, $N = 7$
3 периода замыкания ($\varepsilon = 0.25$ — иллюстрация; уровень: $\varepsilon_7 = 3.66e-03$)



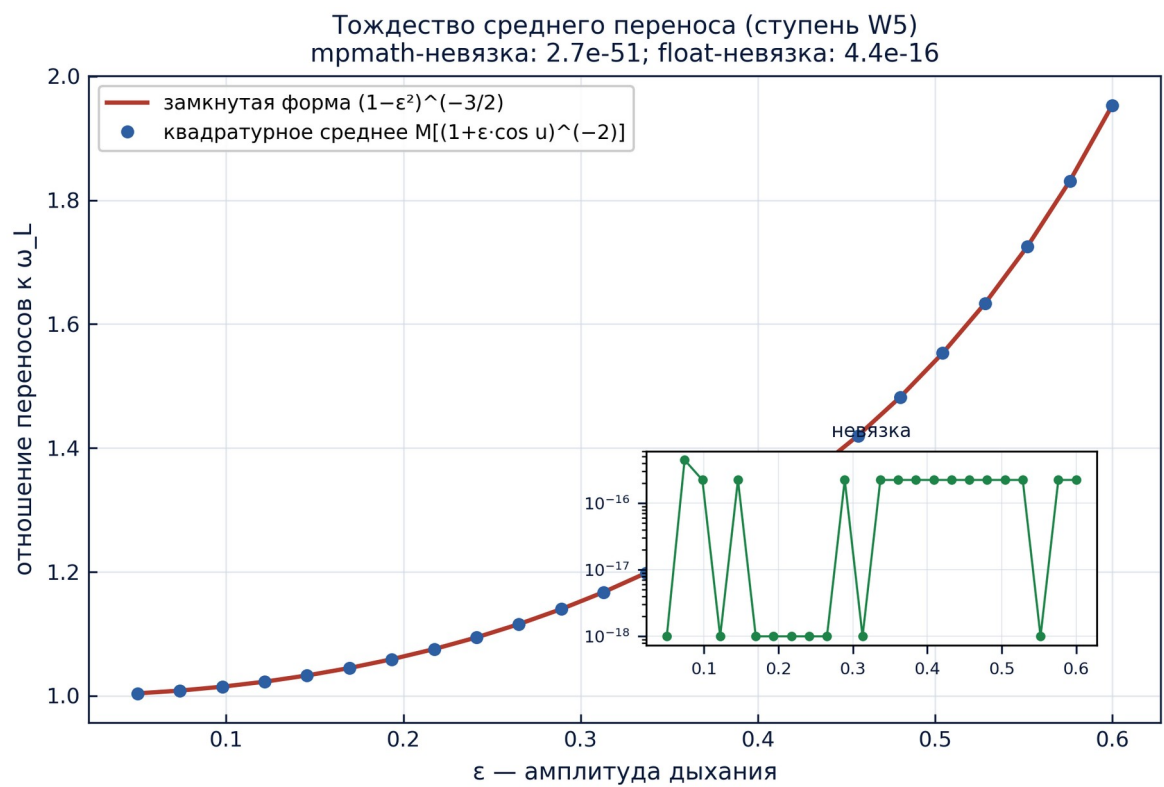
The rosettes and the snapshots of the synchronous program (protocol W6).

4. The numerical certificates

Stage W5: the mpmath residual of the identity $1.0 \cdot 10^{-30}$ over the grid of 16 values $\varepsilon \in [0.05, 0.6]$, the float residual 10^{-14} , the phase-advance error $3.6 \cdot 10^{-13}$. Stage W6: the closure of the level rosettes $N = 7$ at $\varepsilon = \varepsilon_7 = 3.664 \cdot 10^{-3}$ — the residual $1.5 \cdot 10^{-15}$; the shape rigidity along the program — 10^{-16} ; the radial excursion $2R_0\varepsilon$ is nonzero, the breathing is real.

The registers of stages W5–W6 (default preset)

register	tolerance	result
mpmath residual of the identity	1e-30	1.0e-30
float residual of the scan	1e-12	<=1e-14
the 2 pi phase advance	1e-12	3.6e-13
rosette closure (N=7)	1e-9	1.5e-15
shape rigidity	1e-9	1e-16



The quadrature against the closed form over the whole eps grid (protocol W5).